

## Circuit Simulation Lab:27

### *Design of memories*

#### Introduction

The purpose of this experiment is to design a single Port 8 bit RAM Synchronous Read/Write.

#### Software tools Requirement

**Modelsim (Siemens)**

**Xilinx Vivado**

|       |   |                          |
|-------|---|--------------------------|
| Clk   | → | Clock Input              |
| Addr  | → | Address Input            |
| Data  | → | Data bi-directional      |
| c_en  | → | Chip Select              |
| wr_e  | → | Write Enable/Read Enable |
| out_e | → | Output Enable            |

*Design the above Circuit in Verilog HDL and write its testbench and capture the Waveforms,*

*Use Behavior modeling (use parameter to make it generic)*